Advanced Data Management & Laboratory

Examples of exam questions

Exercise
Consider the following information integration specification J = ⟨G,M,S⟩, where:

- Global schema **G** = { **Author**(id, name), **Book**(code, title, authorName) }
- Source schema **S** = { **bestSeller**(bookCode, bookTitle, authorId), **award**(authorName, bookTilte, date) }
- Mapping **M** = { m_1: ∀x,y,z,k,w.**bestSeller**(x, y, z) ∧ **award**(k, y, w) → **Book**(x, y, k),
  m_2: ∀x,y,z,k,w.**bestSeller**(x, y, z) ∧ **award**(k, y, w) → **Author**(z, k) }

- Tell if M is a GAV, LAV, or GLAV mapping.
- Provide the formal definition of retrieved global database for a GAV information integration specification w.r.t. a database.
- Compute the retrieved global database for **J** w.r.t. **D**, where **D** is as follows:
  D = { **bestSeller**(1, “Less”, 10), **bestSeller**(2, “Il cardellino”, 20), **award**(“Greer”, “Less”, 2018),
  award(“Tartt”, “Il cardellino”, 2014), **award**(“Desiati”, “Spatriati”, 2020) }.
- Given the following CQ over G: q = {() | ∃x,y,k,t.Author(x,y) ∧ Book(k,t,y) }, compute cert(q,J,D).
- Write at least two LAV mapping between S and G.

Exercise
Consider the following information integration specification **J** = ⟨G,M,S⟩, where:

- Global schema **G** = { **Researcher**(rid, name), **Paper**(pid, pTitle, authorName) }
- Source schema **S** = { **conferencePaper**(pid, pTitle, rid), **bestPaper**(authorName, pTilte, year) }
- Mapping **M** = { m_1: ∀x,y,z.**conferencePaper**(x, y, z) → ∃w.**Paper**(x, y, w) ∧ **Researcher**(z, w)
  m_2: ∀x,y,z.**bestPaper**(x, y, z) → ∃v.**Paper**(v, y, x) }

- Tell, motivating your answer, if **M** is a GAV, LAV, or GLAV mapping.
- Provide the formal definition of universal solution for an information integration specification w.r.t to a source database **D**.
- Compute a global instance **K** that is an universal solution for **J** w.r.t. **D**, where **D** is as follows:
  D = { **conferencePaper**(1,“title1”,10), **conferencePaper**(2,“title2”,20),
  bestPaper(“Bob”,“title1”,2010),
  bestPaper(“Sam”, “title3”, 2012), **bestPaper**(“Bill”, “title4”, 2021) }.
- Given the following CQ over **G**: **q** = {(y,z) | ∃x.**Paper**(x,y,z)}, compute, motivating your answer, cert(**q**,**J**,**D**).
- Write at least two (possibly new) GAV mapping between **S** and **G**.

Exercise
Consider the following information integration specification **J** = ⟨G,M,S⟩, where:
- Global schema **G** = { **Patient**(pId, pName, pAge), **Hospitalization**(pId, doctor, date) }
- Source schema **S** = { **MediacalRecord**(pId, pName, hospDate, doctoId) }
- Mapping **M** = { m_1: ∀x,y,z,k.**MediacalRecord**(x, y, z, k) → ∃w.**Patient**(x, y, w),
  m_2: ∀x,y,z.**MediacalRecord**(x, y, z, k) → **Hospitalization**(x, k, z)}

- Tell, motivating your answer, if **M** is a GAV, LAV, or GLAV mapping.
- Describe the Naïve Chase Algorithm

- Compute a **ch(M,D)**, where **M** is the above mapping and **D** is as follows:
  D = { **MediacalRecord**(10, “Rossi”, 06/08/2010, 35), **MediacalRecord**(20, “Bianchi”, 09/11/2012, 35), **MediacalRecord**(30, “Verdi”, 15/01/2013, 45) }.
- Given the following CQ over **G**: **q** = {(y,x,w) | ∃z,k. **Hospitalization**(x,y,z) ∧ **Hospitalization** (w,y,k)}, compute the perfect UCQ rewriting of **q** w.r.t. **J**.
- Compute cert(**q**,**J**,**D**).

Exercise

Consider the following information integration specification **J** = ⟨G,M,S⟩, where:

- Global schema **G** = { **Directors**(did, name), **Movies**(mid, title, directorName) }
- Source schema **S** = { **cinemaMovies**(mid, title, did), **oscarWinners**(title, category, directorName) }
- Mapping **M** = { m_1: ∀x,y,z.**cinemaMovies**(x, y, z) → ∃k.**Movies**(x, y, k) ∧ **Directors**(z, k)
  m_2: ∀x,y,z.**oscarWinners** (x, y, z) → ∃w.**Movies**(w, x, z) }

- Tell if **M** is a GAV, LAV, or GLAV mapping motivating your answer.
- Provide the formal definition of universal solution for an information integration specification w.r.t to a source database **D**.
- Compute a global instance **K** that is an universal solution for **J** w.r.t. **D**, where **D** is as follows:
  D = { **cinemaMovies**(m1,“title1”, d1), **cinemaMovies**(m2,“title2”, d2),
  oscarWinners(“title1”,”editing”, “dn1”), **oscarWinners**(““title3”,”editing”, “dn3”),
  oscarWinners (“title4”,”international”, “dn4”) }.
- Given the following CQ over **G**: **q** = {(y,z) | ∃x,w. **Movies**(x, y, z) ∧ **Directors**(w, z)}, compute cert(**q**,**J**,**D**) motivating your answer.
- Write at least two (possibly new) GAV mapping between **S** and **G**.

Exercise

Consider the following information integration specification **J** = ⟨G,M,S⟩, where:

- Global schema **G** = { **Product**(pId, pName, makerName), **Company**(cId, cName, address) }
- Source schema **S** = { **Catalog**(pid, pName, makerId), **Maker**(makerId, makerName) }
- Mapping **M** = { m_1: ∀x,y,z.**Catalog**(x, y, z) → ∃w,t.**Product**(x, y, w) ∧ **Company**(z, w, t)
  m_2: ∀x,y,z.**Maker**(x, y) → ∃z.**Company**(x, y, z) }

- Tell, motivating your answer, if **M** is a GAV, LAV, or GLAV mapping.
- Provide the formal definition of universal solution for an information integration specification w.r.t to a source database D.
- Compute a global instance **K** that is an universal solution for **J** w.r.t. **D**, where **D** is as follows:
  D = { **Catalog**(5,“prod5”,11), **Catalog**(7,“prod7”,22),
  Maker(11,“srl11”), **Maker**(22, “srl22”), **Maker**(33, “srl33”) }.
- Given the following CQ over **G**: **q** = {(k,z) | ∃x.**Product**(x,y,z) ∧ **Company**(k, z, v)}, compute, motivating your answer, cert(**q**,**J**,**D**).
- Write at least one (possibly new) GAV mapping between **S** and **G**.

Exercise

Consider the following DL-Lite TBox T:

T = { doctor ⊑ person,
doctor ⊑ ∃worksFor,
∃worksFor⁻ ⊑ hospital,
medStudent ⊑ doctor,
medStudent ⊑ ¬∃worksFor }

- Tell whether the TBox T is satisfiable, and if so, show a model for T.
- Given the ABox A = {medStudent(bob)}, tell whether the ontology ⟨T,A⟩ is satisfiable (consistent), and if so, show an interpretation that is a model for it.
- Given the following conjunctive query ***q****: { (x) | ∃y.person(x) ∧ worksFor(x,y) ∧ hospital(y) },* compute the perfect rewriting of ***q*** w.r.t. T.
- Given the ABox A' = { doctor(sam), person(bill), worksFor(bill, H3) }, compute the certain answers of the query ***q*** over the ontology ⟨T,A'⟩.
- Add to the TBox T the DL-Lite assertions needed for formalizing the following statement "every doctor has a specialization".

Exercise

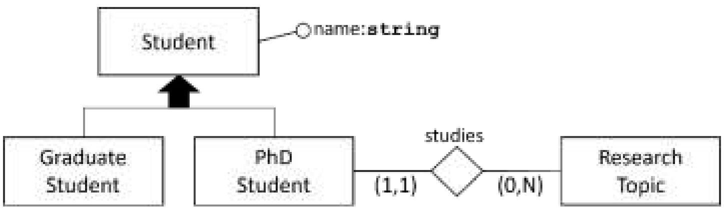
Consider the following TBox T:

T = { book ⊑ product,
e-book ⊑ product,
product ⊑ book ⊔ e-book,
book ⊑ ∃printedBy,
∃printeBy⁻ ⊑ PrintCompany,
e-book ⊑ ¬book }

- Tell, motivating your answer, whether TBox **T** can be expressed in DL-LiteA. If it cannot, write a TBox **T'** containing only the axioms of **T** that are expressible in DL-LiteA.
- Given the ABox **A** = { product(*p1*), book(b1), e-book(eb1), printedBy(*b2, pc1*) }, tell whether the ontology ⟨**T**,**A**⟩ is satisfiable (consistent), and if so, show an *interpretation* that is a *model* for it.
- Given the following conjunctive query ***q****: { (x) | ∃y.product(x) ∧ printedBy(x,y) ∧ PrintCompany(y) },* compute the perfect rewriting of ***q*** w.r.t. **T'**, where **T'** is the DL-LiteA TBox provided by you above.
- Given the ABox **A** above, compute the certain answers of the query ***q*** over the ontology ⟨**T**,**A**⟩.
- Add to the TBox **T** the DL-Lite assertions needed for formalizing the following statement "*every e-book is distributed by at least one e-book store*" (consider to have the role *distributedBy*).

Exercise

Consider the following Entity-Relationship (ER) diagram modelling a portion of the university domain:



- Provide an equivalent formalization with a TBox **T** in Description Logics of the represented domain.
- Tell, motivating your answer, whether the domain can be formalized through a TBox **T'** in DL-LiteA or not. If it cannot, express in DL-LiteA only the portion of the domain that can be captured.
- Given the ABox **A** = { PhdStudent(sam), Student(bill), studies(bill, OBDM) }, tell whether the ontology ⟨**T**,**A**⟩ is satisfiable (consistent), and if so, show an *interpretation* that is a *model* for it.
- Given the following conjunctive query ***q****: { (x) | ∃y.Student(x) ∧ studies(x,y) },* compute the perfect rewriting of ***q*** w.r.t. **T**', where **T**' is the DL-LiteA TBox provided by you above.
- Let **A** be the ABox above, compute the certain answers of the query ***q*** over the ontology ⟨**T'**,**A**⟩.

Exercise

Consider the following domain description about electronic devices:

"We know that both phones and computers are devices, and that a device that is both a phone and a computer is a smartphone. As we also know, a smartphone is both a phone and a computer. Furthermore, we know that each phone has at least one number associated with it".

- Provide a Description Logic TBox **T** that formalizes all the aspects in the domain description.
- Tell, giving motivation for the answer, whether the domain can be formalized through a TBox **T'** in DL-LiteA or not. If it cannot, express in DL-LiteA only the portion of the domain that can be captured.
- Given the ABox **A** = { Phone(d1), Computer(d1) }, tell whether the ontology ⟨**T**,**A**⟩ is satisfiable (consistent), and if so, show an *interpretation* that is a *model* for it.
- Given the following conjunctive query ***q**: { (x) | ∃y.Device(x) ∧ hasNumber(x,y) },* compute the perfect rewriting of ***q*** w.r.t. **T**', where **T**' is the DL-LiteA TBox provided by you above.
- Let **A'** be the ABox { Smatphone(**sm**), Computer(**co**), hasNumber(**de**, **num**) }, compute the certain answers of the query ***q*** over the ontology ⟨**T'**,**A'**⟩.

Exercise

Consider the following TBox T:

T = { car ⊑ vehicle,
bike ⊑ vehicle,
vehicle ⊑ car ⊔ bike,
car ⊑ ∃registeredIn,
$\exists \text{registeredIn}^{-}$ ⊑ country,
bike ⊑ ¬car }

- Tell, motivating your answer, whether TBox **T** can be expressed in DL-LiteA. If it cannot, write a TBox **T'** containing only the axioms of **T** that are expressible in DL-LiteA.
- Given the ABox **A** = { vehicle(*v1*), car(*bmw1*), bike(*bk2*), bike(*bk2, Italy*) }, tell whether the ontology ⟨**T**,**A**⟩ is satisfiable (consistent), and if so, show an *interpretation* that is a *model* for it.
- Given the following conjunctive query ***q**: { (x) | ∃y.vehicle(x) ∧ registredIn(x,y) ∧ country(y) },* compute the perfect rewriting of ***q*** w.r.t. **T'**, where **T'** is the DL-LiteA TBox provided by you above.
- Given the ABox **A** above, compute the certain answers of the query ***q*** over the ontology ⟨**T**,**A**⟩.
- Add to the TBox **T** the DL-Lite assertions needed for formalizing the following statement "*every vehicle has at least one owner*" (consider to have the role *hasOwner*).

Exercise

Let **J** = ⟨G,M,S⟩ be an information integration specification and let **L** be a class of queries. Provide the formal definition of maximally-sound **L**-rewriting of a query **q** w.r.t. **J**.

Exercise

a) Provide the definition of the query containment problem.
b) Given the query **q** = {(y) |∃x,z.S(x,y) ∧ T(y, 'a') ∧ T(y, z)} and **q'** = {(x) |∃y,z. T(x, y) ∧ T(x, z)} , tell, motivating your answer, whether there is a containment relationship between them.

Exercise

c) Provide the definition of the query containment problem.
d) Given the query **q** = {(y) | P(x,y) ∧ P(y, 'c')} and **q'** = {(w) | P(w,k)} , tell, motivating your answer, whether there is a containment relationship between them.

Exercise
Let **Q** be a query language and **L** be an ontology language. Provide the definition of ***Q**-rewritability* in **L**.

Exercise
Let ⟨J,D⟩ be information integration system, where J = ⟨G,M,S⟩. Provide the formal definition of when a global database B satisfies M w.r.t. D in case of sound, complete, and exact mappings semantics.

Exercise
Describe the main motivations behind the development of the key-value model.

Exercise
Describe the advantages of the map-reduce paradigm.

Exercise
Describe the main ideas of the Mondrian algorithm and the requirements guaranteed on the produced output.

Exercise
Describe the principles used for the application of k-anonymity over large data collections.

Exercise
Describe the advantages of the multidimensional model.

Exercise
Describe the use of Randomized response to support differential privacy.